

diagnosis and management of this condition. In the absence of necrosis, homogeneous enhancement occurs after infusion of water-soluble contrast media in pituitary adenomas. In the cases presented here, peripheral or ring-like enhancement with a lucent center suggested a necrotic adenoma. Rupture of a cystic craniopharyngioma might possibly give similar clinical and CT findings, but skull x-ray films should differentiate the two. Although very rare, subarachnoid bleeding from a large aneurysm of the internal carotid artery eroding into the sella may mimic pituitary apoplexy and should be excluded by

angiography.

If the hemorrhagic pituitary tumor has no suprasellar extension but only enlarges the sella or extends inferiorly into the sphenoid sinus, in the majority of cases only the plain skull x-ray films or tomograms of the sella will alert one to the diagnosis in this catastrophic event. Thus, skull x-ray films in this entity are still the most useful and easily obtainable examination. However, with thin coronal cuts and improved resolution, localized intrasellar pathology can sometimes be seen on CT scanning and certainly will be routine in the very near future.

## References

1. Dawson BH, Kothandaram P: Acute massive infarction of pituitary adenomas. *J Neurosurg* 37:275-279, 1972.
2. McDonald TJ, Tolis G, McGarry EE: Twenty-two cases of pituitary apoplexy. *Ann R Coll Phys Surg Can* 8:19, 1975.
3. Conomy JP, Ferguson JH, Brodkey JS, et al: Spontaneous infarction in pituitary tumours: Neurologic and therapeutic aspects. *Neurology* 25:580-587, 1975.
4. Rovitt RL, Fein JM: Pituitary apoplexy: A review and reappraisal. *J Neurosurg* 37:280-288, 1972.
5. Sachdev Y, Evered DC, Hall R: Spontaneous pituitary necrosis. *Br Med J* 1:942, 1976.
6. David NJ, Glaser JS, Gargano FP: Pituitary apoplexy in clinical perspective, in Glaser JS, Smith JL (eds): *Neuro-ophthalmology: Symposium of the University of Miami and the Bascom Palmer Eye Institute*. St Louis, CV Mosby Co, 1975, vol 8, pp 140-165.

## The Effects of Bicarbonate on Blood Coagulation

Dennis W. Wong, PharmD; Fred S. Mishkin, MD; Toshiyuki T. Tanaka, MD

• The anticoagulant effects of sodium and potassium bicarbonate were investigated with use of fresh human whole blood obtained from normal healthy volunteers. Results from prothrombin and thrombin clotting time determinations demonstrated that bicarbonate can interfere with the clotting process. Clinical data obtained from patients who received intravenous administration of sodium bicarbonate supported these findings.

(*JAMA* 244:61-62, 1980)

WHILE investigating bicarbonate as a buffer in a radioisotope labeling procedure for fibrinogen, we found prolonged in vitro clotting times in the buffered fibrinogen solution. Since earlier work contained no information dealing directly with the effect of bicarbonate on clotting,<sup>1-10</sup> we examined in vitro the effect on the prothrombin and thrombin clotting times due to addition of sodium or potassium bicarbonate to citrated blood. We tried to separate this effect from changes due to pH and ionic strength. Since we found coagulation

changes occurred at concentrations that could be expected with massive infusions of bicarbonate, we sought confirmation of our in vitro findings in patients being treated for acute acidosis.

## Methods

In vitro studies were performed using fresh blood samples collected from healthy volunteers by venipuncture. Prior to addition of the whole blood, a predetermined amount of NaHCO<sub>3</sub> solution (7.5%, pH 7.8), ranging from 1 to 12 mg in less than 0.1 mL volume, was added to the specimen test tubes used for blood collection, each containing 0.3 mL of 3.8%, 0.13M trisodium citrate solution. When the required amount of NaHCO<sub>3</sub> to be added exceeded a 0.1 mL volume, a weighed amount of reagent grade powder was dissolved in the citrate solution. Exactly 2.7 mL of the fresh whole blood was immediately added to the sample test tube and well mixed,

giving a ratio of one part citrate solution to nine parts of test sample. The quantity of NaHCO<sub>3</sub> added to the blood samples was equivalent to administering 1 to 10 mEq/kg of body weight to a 70-kg adult, assuming uniform distribution only in the blood. The plasma ionic strength was assumed to equal normal saline (0.9% NaCl solution) of 0.154.<sup>4,10</sup>

Plasma was separated from the whole elements by centrifugation at 2,000 rpm (540 g) for 30 minutes. One-stage prothrombin time (PT), by the addition of thromboplastin and Ca<sup>++</sup> ions to 0.2 mL of plasma, was determined according to standard laboratory methods. The same plasma sample was diluted to 1:10 with Owren's pH 7.35 sodium barbital and sodium chloride buffer. Thrombin clotting time was determined by adding 0.1 mL (10 units/mL) of topical thrombin to 0.1 mL of a plasma sample previously incubated at 37 °C for two to five minutes and mechanically recorded using a clot timer. Blood sample pH was determined at 37 °C, using a pH meter with a combination glass and ceramic electrode prior to plasma separation. Control plasma samples were tested without the addition of NaHCO<sub>3</sub>. Similar studies were performed using reagent-grade KHCO<sub>3</sub> and NaCl powder.

Data from patients receiving large infusions of bicarbonate for acute acidosis were analyzed if the tests ordered included arterial blood pH, P<sub>O<sub>2</sub></sub>, P<sub>CO<sub>2</sub></sub>, and plasma HCO<sub>3</sub> levels, PT, partial thromboplastin time (PTT), and thrombin clotting time. Laboratory results from blood samples prior to intravenous (IV) administration of NaHCO<sub>3</sub> served as baseline controls. The second blood sample had to be obtained from the patient within 12 to 15 minutes following each course of NaHCO<sub>3</sub> therapy to allow complete equilibration of

From the Division of Nuclear Medicine, Department of Radiology, Charles E. Drew Postgraduate Medical School; and the Martin Luther King, Jr. General Hospital of Los Angeles County, Los Angeles.

Reprint requests to Department of Radiology, Martin Luther King, Jr. General Hospital, 12021 S Wilmington Ave, Los Angeles, CA 90059 (Dr Wong).

# Correlation of Change in Prothrombin Time With Bicarbonate Infusion in Acute Acidosis

| Patient                              | Before Bicarbonate Therapy |                                        |                            |                             |                |                | After Bicarbonate Therapy                               |             |                                        |                            |                             |                |                |
|--------------------------------------|----------------------------|----------------------------------------|----------------------------|-----------------------------|----------------|----------------|---------------------------------------------------------|-------------|----------------------------------------|----------------------------|-----------------------------|----------------|----------------|
|                                      | Blood<br>pH                | HCO <sub>3</sub> <sup>-</sup> ,<br>mEq | Po <sub>2</sub> ,<br>mm Hg | Pco <sub>2</sub> ,<br>mm Hg | PT,<br>s*      | PTT,<br>s*     | Total HCO <sub>3</sub> <sup>-</sup><br>Injected,<br>mEq | Blood<br>pH | HCO <sub>3</sub> <sup>-</sup> ,<br>mEq | Po <sub>2</sub> ,<br>mm Hg | Pco <sub>2</sub> ,<br>mm Hg | PT,<br>s*      | PTT,<br>s*     |
|                                      |                            |                                        |                            |                             |                |                |                                                         |             |                                        |                            |                             |                |                |
| 1 (severe metabolic acidosis)        | 6.76                       | 7                                      | 68                         | 55                          | 10.7<br>(10.6) | 35.5<br>(29.7) | 223.0†                                                  | 7.39        | 15                                     | 140                        | 20                          | 20.2<br>(10.6) | 54.3<br>(29.2) |
| 2 (cardiopulmonary arrest)           | 7.40                       | 17                                     | 28                         | 96                          | 14.3<br>(10.6) | 83.4<br>(32.4) | 89.2                                                    | 7.23        | 20                                     | 52                         | 74                          | 16.1<br>(10.5) | 93.8<br>(32.4) |
| 3 (diabetic ketoacidosis)            | 7.02                       | <5                                     | 65                         | 18                          | 12.2<br>(10.6) | 38.2<br>(36.0) | 89.2                                                    | 7.27        | 15                                     | 382                        | 32                          | 22.9<br>(10.6) | 58.4<br>(35.5) |
| 4 (septic shock and lactic acidosis) | 7.14                       | <8                                     | 83                         | 20                          | 13.0<br>(10.6) | 38.8<br>(34.9) | 89.2                                                    | 6.98        | 8                                      | 136                        | 19.3                        | 28.2<br>(10.0) | 47.3<br>(32.0) |

\*Normal values for prothrombin time (PT) and partial thromboplastin time (PTT) are given in parentheses.

†Blood samples were collected 15 minutes after intravenous infusion—a total of 223 mEq in 11 hours.

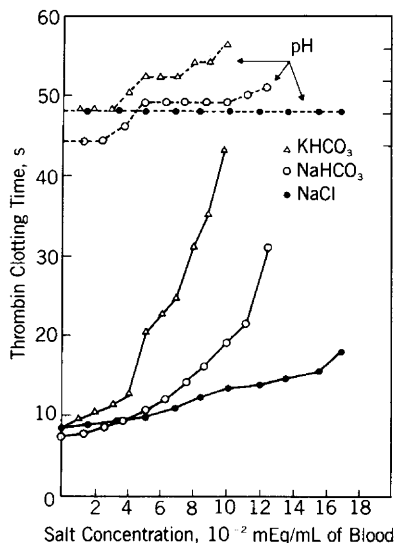


Fig 1.—Effect of addition of KHCO<sub>3</sub>, NaHCO<sub>3</sub>, or NaCl on thrombin clotting time and pH of citrated whole blood. All salts increased thrombin clotting time, but NaCl did not increase pH. For bicarbonate salts, pH and thrombin clotting time rose together, more sharply than with addition of NaCl.

NaHCO<sub>3</sub> in the blood. All except patient 1 received bolus injection of bicarbonate; preinjection and postinjection bicarbonate samples were therefore separated by approximately 15 minutes. In patient 1, bicarbonate was given by IV infusion over an 11-hour period; the control and postinfusion bicarbonate samples were therefore separated by 11 hours. A total of four patients satisfied these criteria.

## Results

Figure 1 summarizes the in vitro thrombin clotting times of an average of three determinations for blood containing either KHCO<sub>3</sub>, NaHCO<sub>3</sub>, or NaCl. Sodium chloride was used to demonstrate the effect of alteration of ionic strength on the clotting process. At the salt concentrations used, the ionic strength, which depends on molar concentration of the solutions, ranged from 0.154 to 0.254 for KHCO<sub>3</sub>,

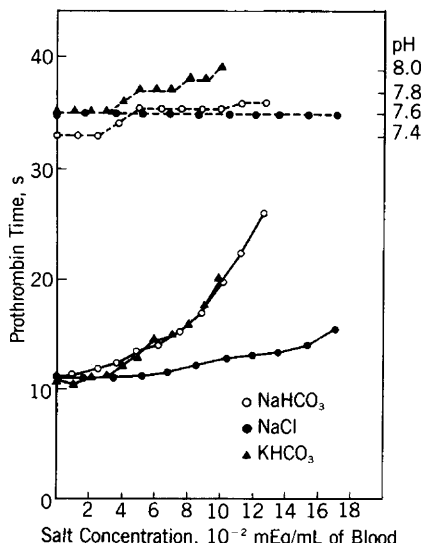


Fig 2.—Effect of addition of KHCO<sub>3</sub>, NaHCO<sub>3</sub>, or NaCl on prothrombin time of normal human blood samples. All salts prolonged prothrombin time, yielding decreased effective percentage of prothrombin.

from 0.154 to 0.279 for NaHCO<sub>3</sub>, and from 0.154 to 0.325 for NaCl. The pH rose for both KHCO<sub>3</sub> and NaHCO<sub>3</sub>, while it remained virtually constant with the addition of NaCl. All the salts inhibited the conversion of fibrinogen to fibrin, the potassium salt more markedly than the sodium salt of bicarbonate. Figure 2 shows a similar effect on the PT.

The Table summarizes the patient data. Although many variables affect the clotting process in vivo, over the short period of time these observations were made, the most striking consistent changes were in Po<sub>2</sub> and bicarbonate concentrations in response to administration of oxygen and massive IV doses of bicarbonate. In each case, the PT became prolonged in comparison with the baseline value.

## Comment

The experimental results and patient data suggest that bicarbonate can interfere with the clotting process; perhaps the reason for this is related to both its effect on blood pH and ionic strength as well as the salt itself. Such an effect may become important in clinical situations that require administration of large doses of bicarbonate over a short period of time. Although we studied only adults, perhaps the most susceptible group would be small infants. Physicians responsible for administering large doses of bicarbonate should be aware of its potential effect on the clotting process.

Morris Dichter provided technical assistance for the prothrombin and thrombin clotting time determinations.

## References

1. Abildgaard U: Acceleration of fibrin polymerization by acetate and other low molecular ions. *Scand J Clin Lab Invest* 16:521-530, 1964.
2. Astrup T: Biochemistry of blood coagulation. *Acta Physiol Scand* 19:22-52, 1944.
3. Briggs R, Denson KEW: Natural and pathological inhibitors of blood coagulation, in Briggs R (ed): *Human Blood Coagulation, Hemostasis and Thrombosis*. Oxford, England, Blackwell Scientific Publications, 1972, pp 133-135.
4. Edsall JT, Lever WF: Effects of ions and neutral molecules on fibrin clotting. *J Biol Chem* 191:735-756, 1951.
5. Harmison CR, Mammen EF: Molecular characteristics of substances active in blood coagulation, in Seegers WH (ed): *Blood Clotting Enzymology*. New York, Academic Press Inc, 1967, pp 24-101.
6. Lovelock JE, Porterfield BM: Blood clotting: The function of electrolytes and of calcium. *Biochem J* 50:415-420, 1952.
7. Ratnoff OD, Potts AM: The accelerating effect of calcium and other cations on the conversion of fibrinogen to fibrin. *J Clin Invest* 33:206-210, 1954.
8. Robbins KC: A study on the conversion of fibrinogen to fibrin. *Am J Physiol* 142:581-588, 1944.
9. Seegers WH: Nature of thrombin, in Seegers WH (ed): *Prothrombin*. Boston, Harvard University Press, 1962, pp 382-443.
10. Stormorken H, Newcomb TF: The use of buffers in blood coagulation studies. *Scand J Clin Lab Invest* 8:237-242, 1956.